



**KTH Microelectronics
and Information Technology**

Exam in ID2209 Distributed Artificial Intelligence and Intelligent Agents, 2024-01-10, 08:00-12:00

Rules

This exam is “closed book” and you are not allowed to bring any material or equipment (such as laptops, PDAs, or mobile phones) with you. The only exceptions are an English to “your favorite language” dictionary and pencils.

Instructions

- Please read the entire exam first!
- Write clearly
- Each sheet of paper must contain your name, ”personnummer”, Problem number and a unique sheet number
- Write only on one page of a sheet. Do not use the back side
- Only one Problem must be reported on each sheet
- If more than one sheet is needed the continuation should be clearly noted on the beginning of each sheet and the sheet numbers used should be consecutive
- Always motivate your answers. Lack of clearly stated motivation can lead to a reduction in the number of points given
- The tasks are not necessarily sorted in order of difficulty. If you get stuck it might be a good idea to go on to the next task.
- The Parts are sorted in order of difficulty but inside of the Parts problems are not sorted in the order of difficulty

Grading

1. In order to pass exam (grade E) you have to get at least 36 points from questions from Part I (this includes your bonus points).
2. In order to get grade D you have to pass exam (see item 1) and get 49 points from all questions (this includes your bonus points).
3. In order to get grade C you have to pass exam (see item 1) and get 63 points from all questions (this includes your bonus points).
4. In order to get grade B you have to pass exam (see item 1) and get 74 points from all questions (this includes your bonus points).
5. In order to get grade A you have to pass exam (see item 1) and get 85 points from all questions (this includes your bonus points).

GOOD LUCK!

Part I (48 p.)

What is an agent?

1. Let us imagine a smart home where most of devices (such as refrigerator, washing machine, dish-washing machine, even wardrobe etc.) are software agents (at least they are represented by software agents). Which of agent's properties should these agents employ and which of agent's properties they can't employ (if any)? Illustrate your answer by examples.

(6p)

2. What is difference between utility functions for cooperative and for self-interested agents?

(3p)

Agent theory

3. When is it useful consider a system (like agents) as intentional systems and when it doesn't make sense?

(4p)

4. Explain duality of operators of a Modal logic.

(4p)

Agent Architectures

5. What is similarity and what are differences between subsumption architecture and layered hybrid agent architecture?

(4p)

Negotiation

6. What is the Majority Graph? How it is used to check a winner in voting protocol?

(4p)

7. What are Strictly Competitive and zero-sum interactions? Give examples. How often they are met in real life?

(4p)

8. If we have the following payoff matrix below representing a negotiations. What is/are pure Nash equilibrium(s) for such negotiation? Explain your answer.

		Defect	Coop
j	Defect	1 1	2 0
	Coop	0 2	3 3

(4p)

Communication

9. Explain what does the following KQML text mean:

(forward

```
:from Agent1
:to Agent2
:sender Agent1
:receiver Agent3
:language KQML
:ontology kqml-ontology
:content (tell
  :sender Agent1
  :receiver Agent2
  :language KIF
  :ontology Blocks-World
  :content (On (Block A) (Block B))))?)
```

(4p)

Coordination

10. What are main problems with organizational structure as a coordination technique?

(3p)

11. What are advantages and disadvantages of Mutual Modelling as a coordination technique?

(4p)

Mobile agents

12. Explain difference between mobile agents and remote procedure call.

(4p)

Part II (27 p.)

13. What are 3 main Isaac Asimov's Laws of robotics? Why the order of application of Asimov's laws is important?

(5p)

14. Given truthful (on Figure 1) and untruthful (on Figure 2) rankings for buying memory by CS, ECEn and ME department calculate the tax to be paid by CS in each of these cases. Explain your answer.

i	$v_i(g)$		
	16TB	32TB	64TB
CS	1	2	3
ECEn	3	1	2
ME	3.5	1	2
$\sum_i v_i(g)$	7.5	4	7

Figure 1

i	$v_i(g)$		
	16TB	32TB	64TB
CS	1	2	4
ECEn	3	1	2
ME	3.5	1	2
$\sum_i v_i(g)$	7.5	4	8

Figure 2

15. Explain Distributed Planning for Centralized plans. Give examples.

(5p)

16. Explain what does the following statement in KIF mean:

($\Rightarrow$ (believes John ?p) (believes Mary ?p))

(5p)

17. Let us consider a Muddy Children puzzle (see Appendix) with n children. How many rounds is needed for a child to understand that his/her forehead is muddy for the case of 1,2 and 3 muddy children? Explain your answer. Just giving a number without explanation will not be considered as a correct answer.

(6p)

Part III (22 p.)

18. What is self-organization and how it is different from emergency?

(7p)

19. Explain what is Condorcet winner and Slater ranking? Illustrate them using Majority graphs.

(8p)

20. What are problems if we make a naïve translation of the statement “Janine believes Cronos is father of Zeus” into the classical logical statement “*believes(janine, father(zeus,cronos))*” in case of an intentional system.

(7p)

Appendix:

There is a large group of children playing in the garden (their perceptiveness, truthfulness, and intelligence being common knowledge, it goes without saying). A certain number of children (say k) get mud on their foreheads. Each child can see the mud on others but not on his own forehead. If $k > 1$ then each child can see another with mud on its forehead, so each one knows that at least one in the group is muddy.

The father first announces that at least one of them is muddy (which is something they know already); and then he repeatedly asks them 'Does any of you know whether you have mud on your own forehead?' If children reason with available knowledge they are able to understand if they are muddy.

-----End of Exam-----